

Impact Statement

This project builds the retrieval component of a Retrieval-Augmented Generation (RAG) system designed to help smallholder farmers and agricultural extension workers in Sub-Saharan Africa quickly find trustworthy answers to crop, pest, soil, and climate-adaptation questions. The core problem it addresses: relevant agricultural knowledge already exists in extension manuals and technical guides, but is often too slow or difficult to locate at the moment a farmer needs it, delaying decisions that affect yield, income, and land health.

Positive impacts: Faster, more accurate retrieval means farmers and extension workers spend less time searching and more time acting, identifying a disease correctly before it spreads, or applying fertilizer at the right stage rather than guessing. Our pipeline improved retrieval accuracy substantially over a naive keyword baseline (nDCG@5 rose from 0.551 to 0.836, specifically by resolving cases where lexically similar but wrong-intent documents (e.g. "prevention" advice retrieved for a "management" question) would otherwise mislead a user. That improvement translates directly into fewer inappropriate recommendations reaching farmers.

Risks and harms: The clearest risk is a false sense of completeness: our corpus audit found that Maize, Tomato, Rice, and Cassava make up over half the documents, while staple crops like Pearl millet are barely represented, and just 4 countries account for 58% of coverage. A farmer growing an underrepresented crop in an underrepresented region may receive generic or poorly matched advice while the system still presents an answer with apparent confidence, a subtler harm than an obvious "no results" failure. A second risk: 91.7% of documents are synthetically generated rather than sourced from a real, checkable publication, which could overstate the system's authority if that distinction isn't surfaced to users.

Values: The project is built around transparency (every document carries a source attribution, distinguishing grounded from synthetic content) and reliability (retrieval quality is measured against graded expert relevance judgments, not just keyword overlap). We treat accessibility and equity as values still owed to this community rather than fully delivered, the current system is English-only and text-only, a real limitation for the women farmers and lower-literacy users named in our original problem statement.

Mitigation: We plan to disclose grounded-vs-synthetic provenance in any user-facing answer, prioritize sourcing for underrepresented crops/regions before wider rollout, and treat local-language and voice support as required next steps.

Limitations: Our evaluation relies on a fixed, English-language query set that may not reflect how real farmers phrase questions; performance on real-world, unscripted queries remains untested.